



NTT Data & IST ML Challenge 2025/26

Portuguese Fake News Detection

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72%

of Portuguese people say that they don't know what's real
or fake on the Internet^[1]

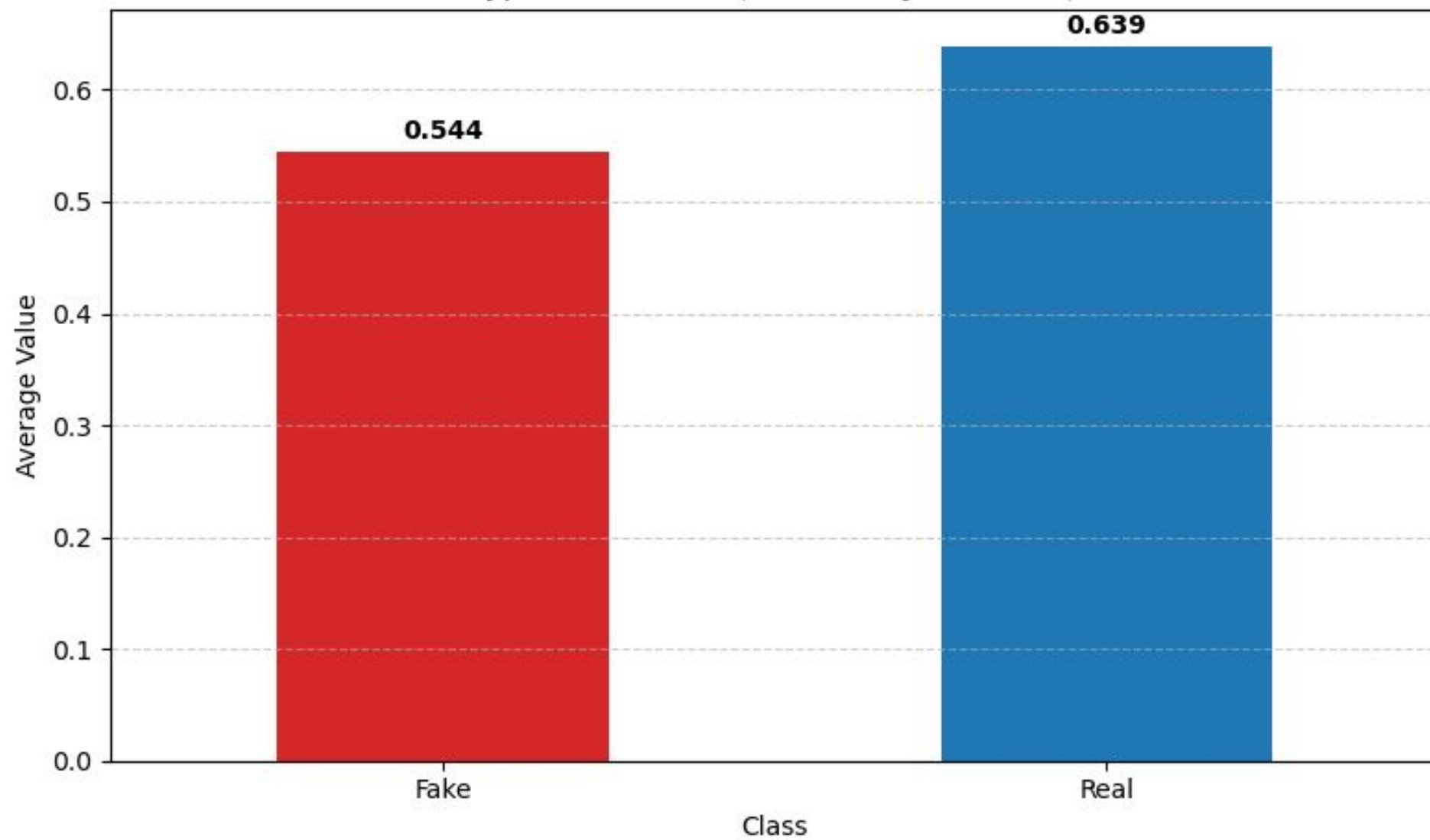
[1] : <https://www.sulinformacao.pt/en/2024/06/desinformacao-mais-detetada-pelos-portugueses-e-sobre-politica/>

Dataset Overview

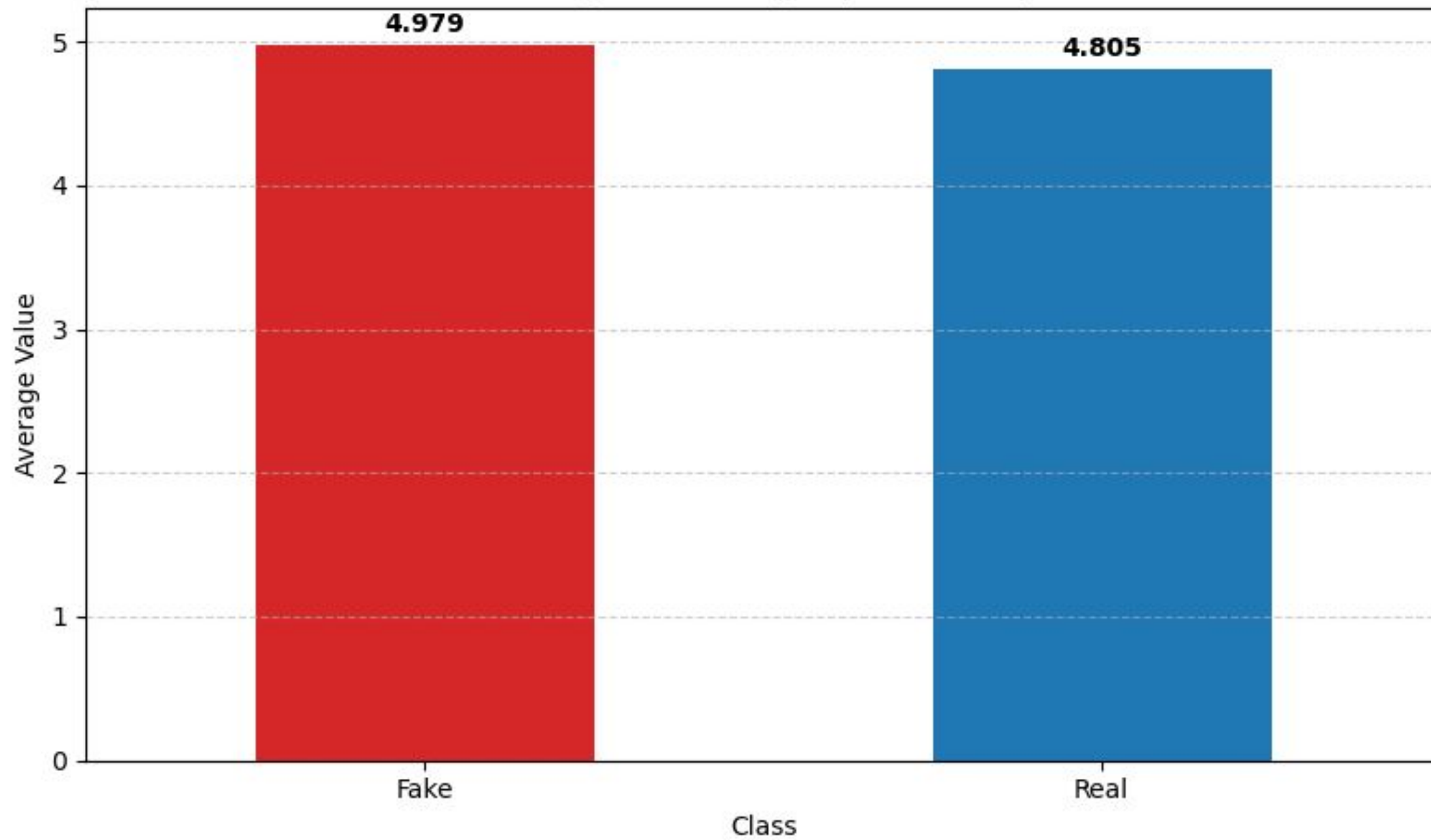
- FakeNews-PT dataset from the fake-news-pt-eu repository
- 60.000+ news articles
- Separated into train, validation, test sets
- Binary classification

Set	Fake News	Real News
Training Set	25209	25378
Validation Set	3141	3180
Test Set	3169	3155

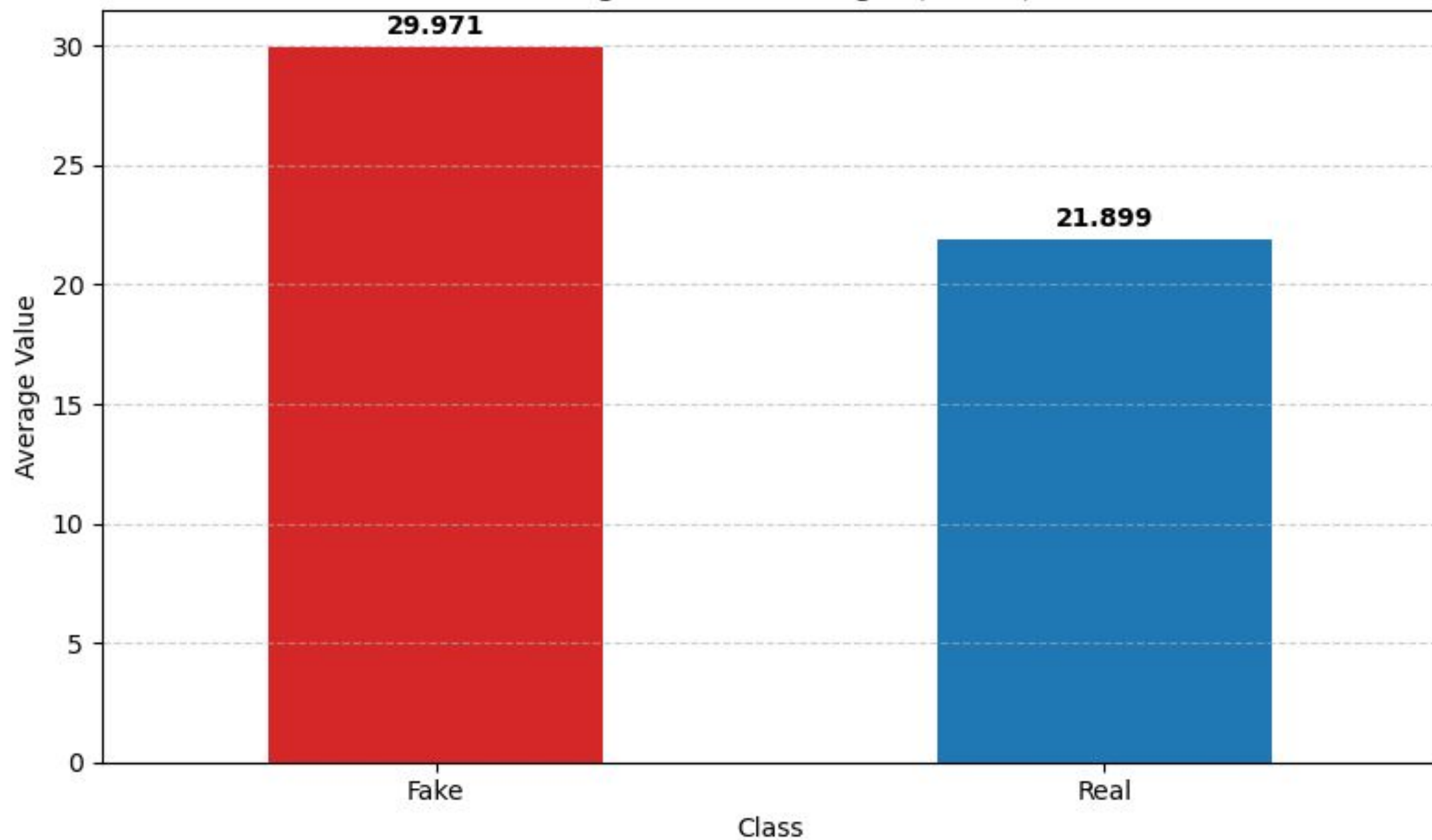
Type-Token Ratio (Vocabulary Richness)



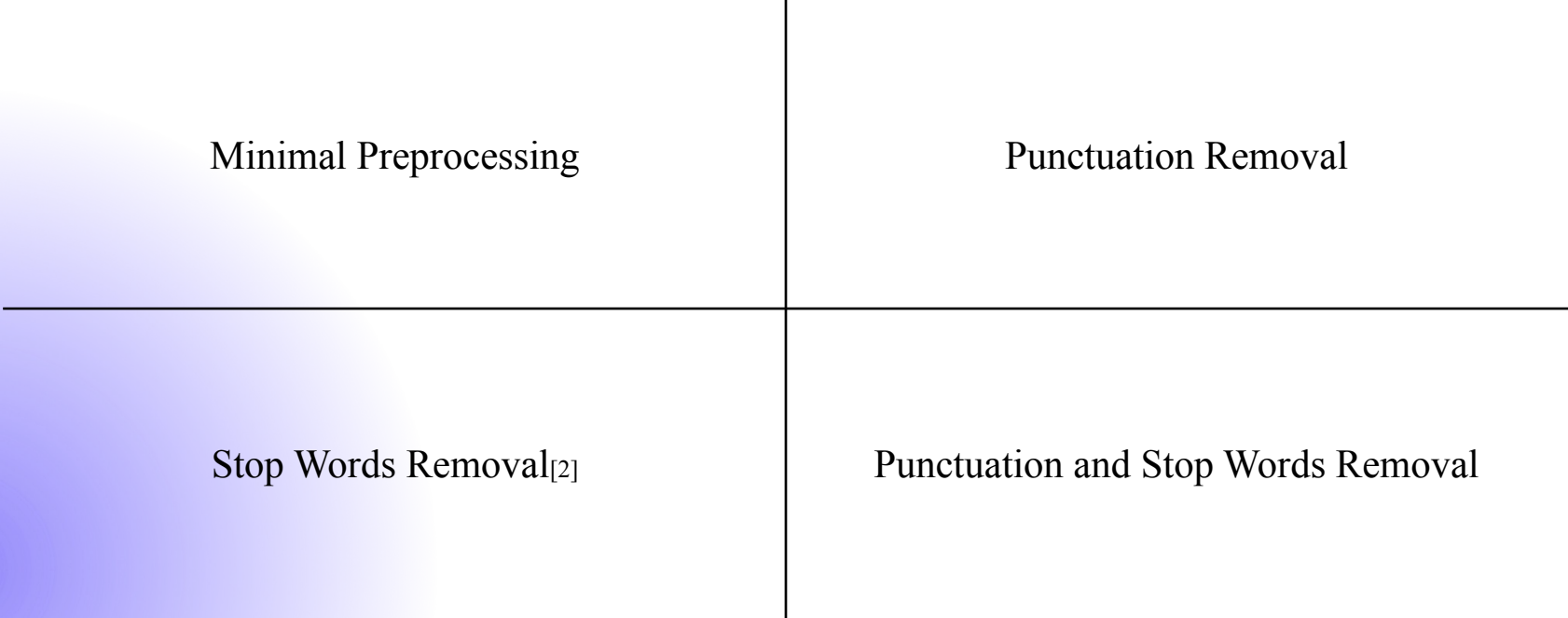
Average Word Length (Characters)



Average Sentence Length (Words)



Text Preprocessing Strategies



Minimal Preprocessing

Punctuation Removal

Stop Words Removal[2]

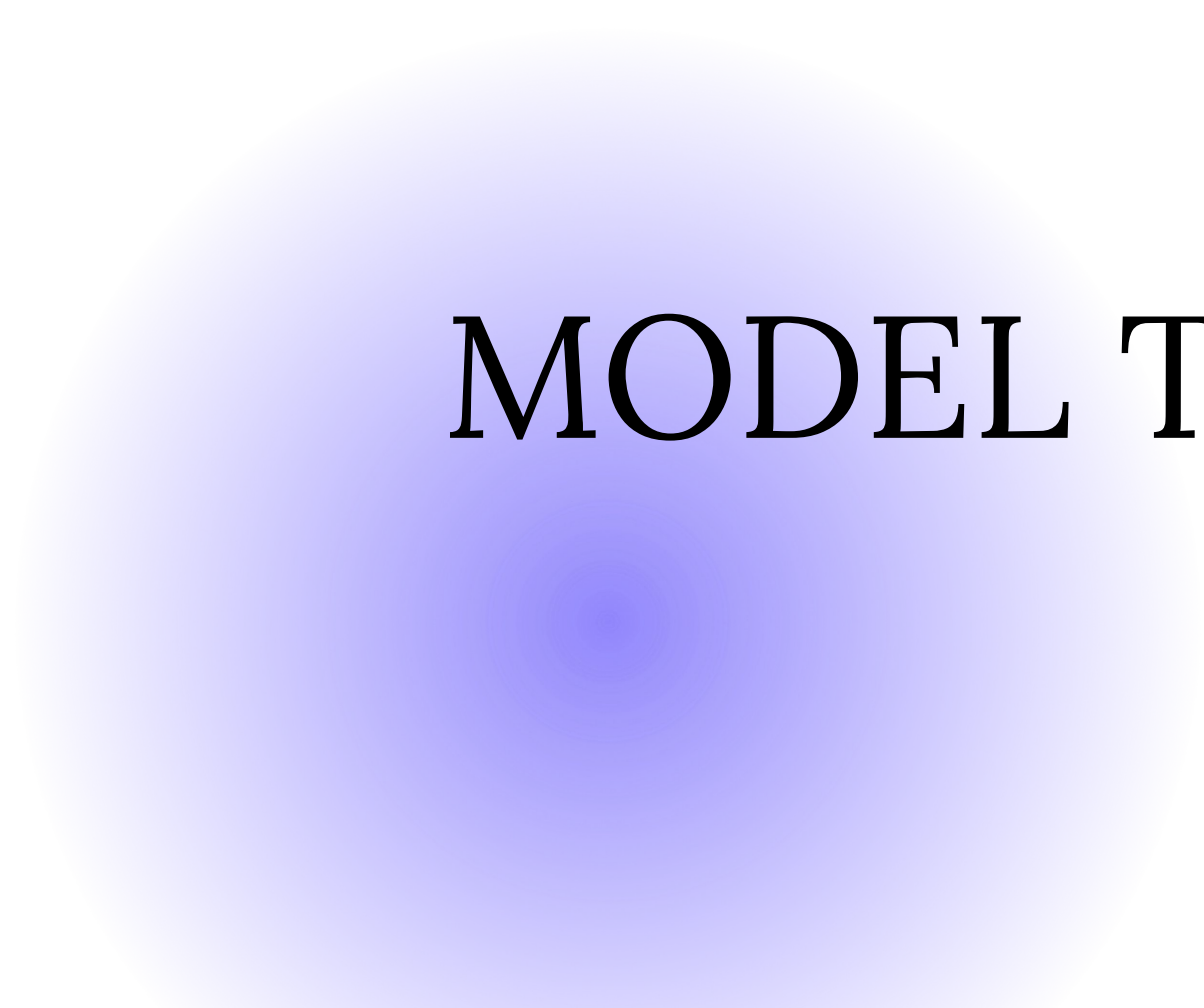
Punctuation and Stop Words Removal

[2] : Portuguese Stop Words from github.com/nltk/nltk_data

Feature Extraction

Usage of TF-IDF with:

- Top 5000 features by term frequency
- Minimum (10) and Maximum (90%) document frequency
- IDF smoothing



MODEL TRAINING

Model	Hyperparameters	Best Configuration
Decision Tree	- max_depth: [10, 20, 50, None] - min_samples_split: [2, 5, 10]	{max_depth: 50, min_samples_split: 2}
Gaussian Naive Bayes	- var_smoothing: [1e-9, 1e-8, 1e-7, 1e-6]	{var_smoothing: 1e-6}
Logistic Regression (L2 Regularization)	- C: [0.01, 0.1, 1, 10, 100] - solver: ['liblinear', 'saga']	{C: 10, solver: 'saga'}
Logistic Regression (L1 Regularization)	- C: [0.01, 0.1, 1, 10] - solver: ['liblinear', 'saga']	{C: 1, solver: 'saga'}
Multi-Layer Perceptron	- hidden_layer_sizes: [(128,), (256,), (128,64), (256,128)] - alpha: [0.0001, 0.001, 0.01] - learning_rate_init: [0.0001, 0.001]	{hidden_layer_sizes: (256, 128), alpha: 0.01, learning_rate_init: 0.001}

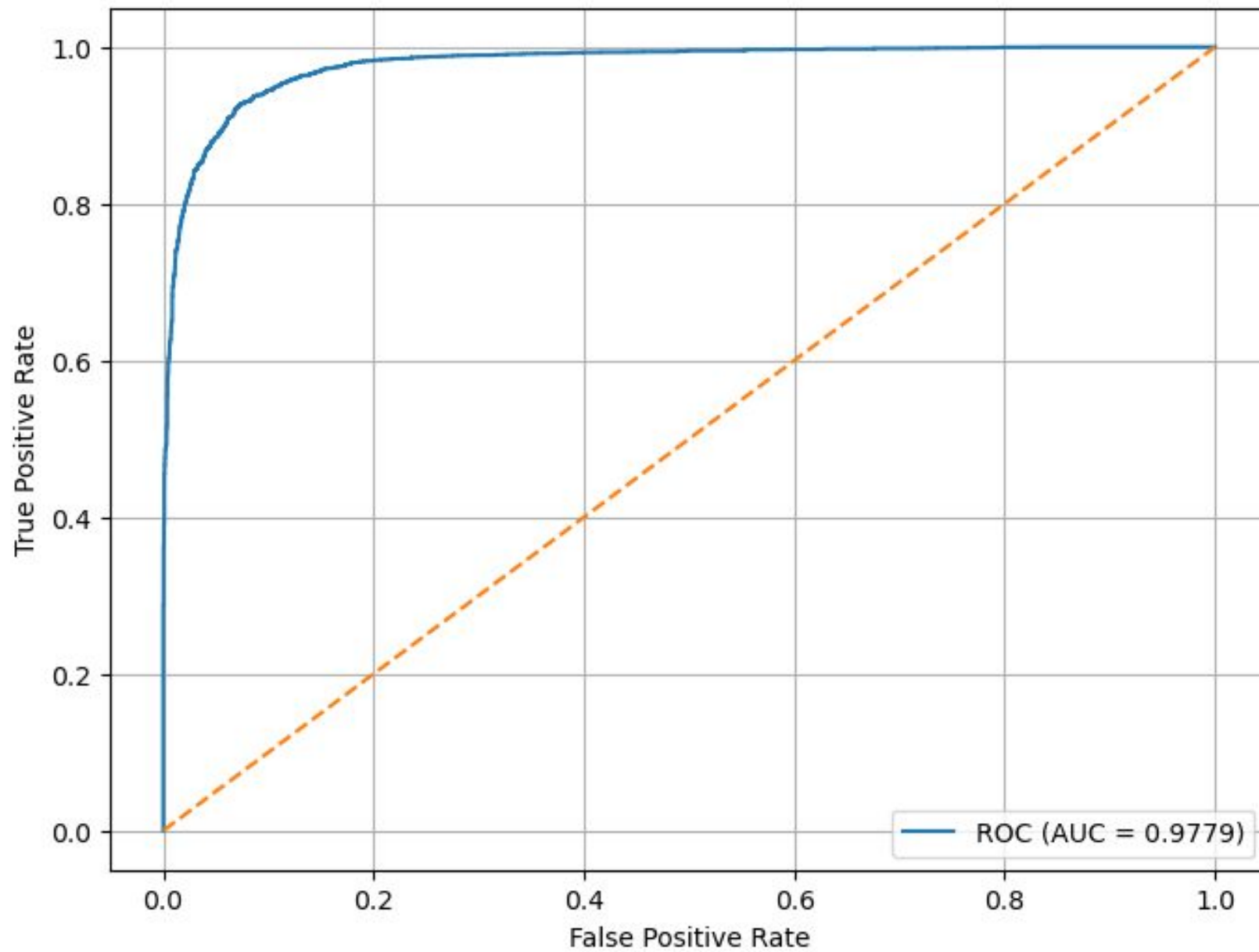
Preprocessing Strategies Analysis

Strategy	Accuracy	Precision (Macro)	Recall (Macro)	F1 (Macro)
Minimal Preprocessing	0.8869	0.8869	0.8869	0.8869
Punctuation Removal	0.8058	0.8073	0.8058	0.8055
Stop Words Removal	0.8875	0.8876	0.8875	0.8875
Punctuation and Stop Words Removal	0.8855	0.8857	0.8855	0.8855

Model Comparison

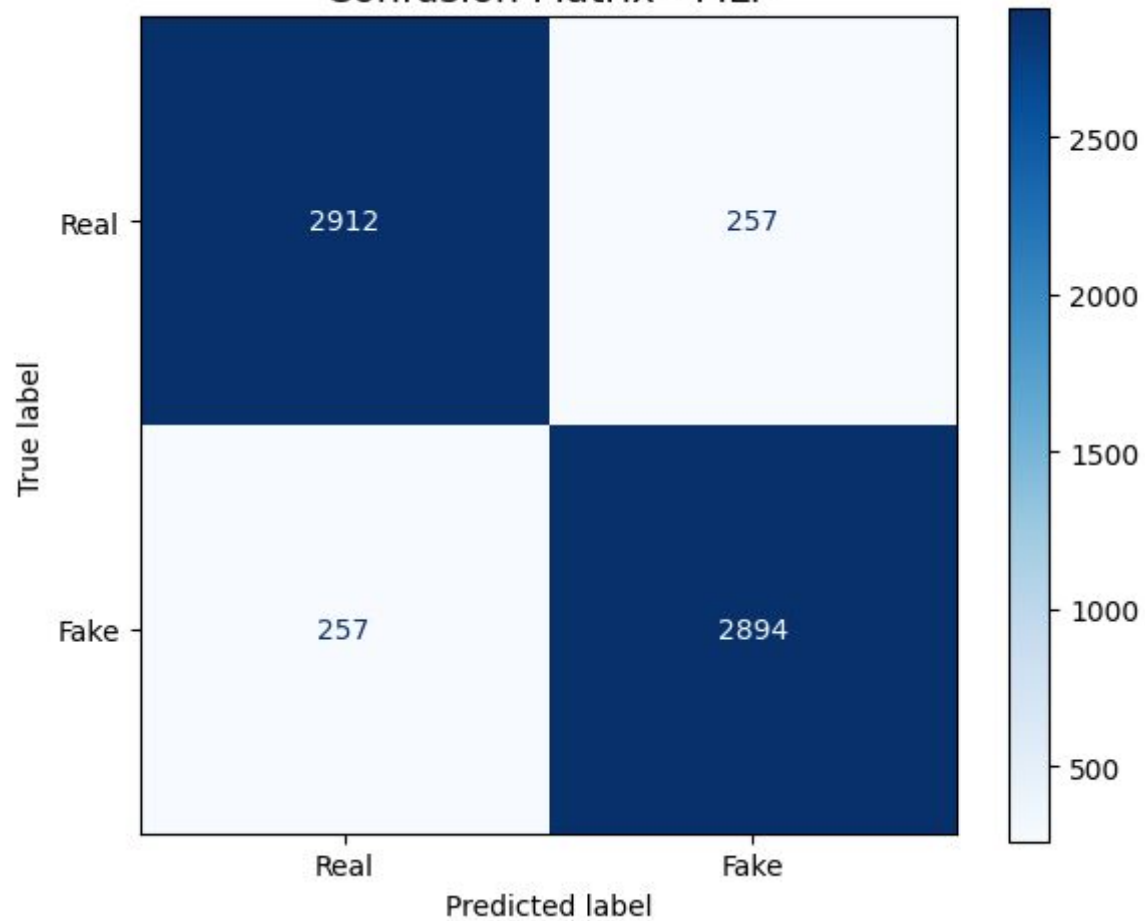
Model	Accuracy	Precision (Macro)	Recall (Macro)	F1 (Macro)
Decision Tree	0.8589	0.8592	0.8588	0.8588
Gaussian Naive Bayes	0.8516	0.8517	0.8516	0.8516
Logistic Regression L2	0.9000	0.9000	0.9000	0.9000
Logistic Regression L1	0.9009	0.9010	0.9009	0.9009
MLP	0.9261	0.9262	0.9261	0.9261

ROC Curve - MLP

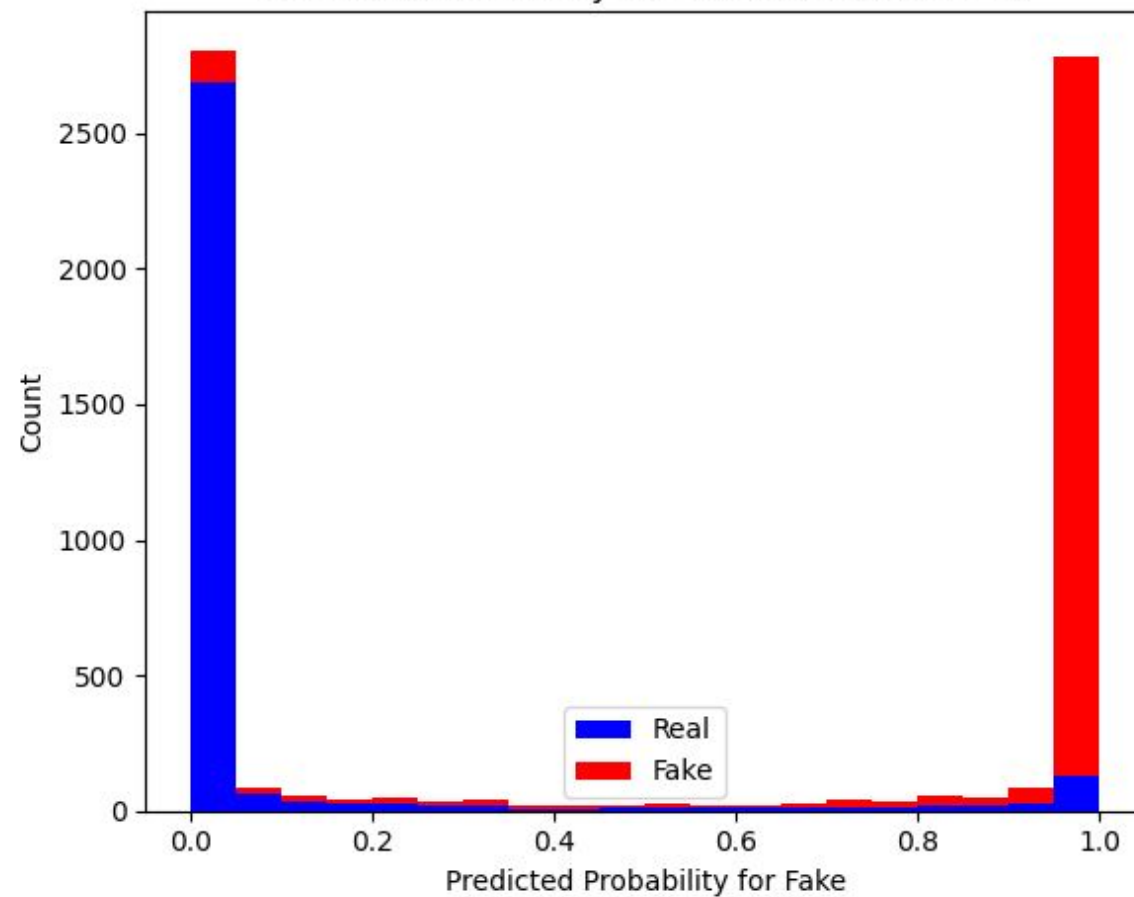


AUC = 0.9779

Confusion Matrix - MLP

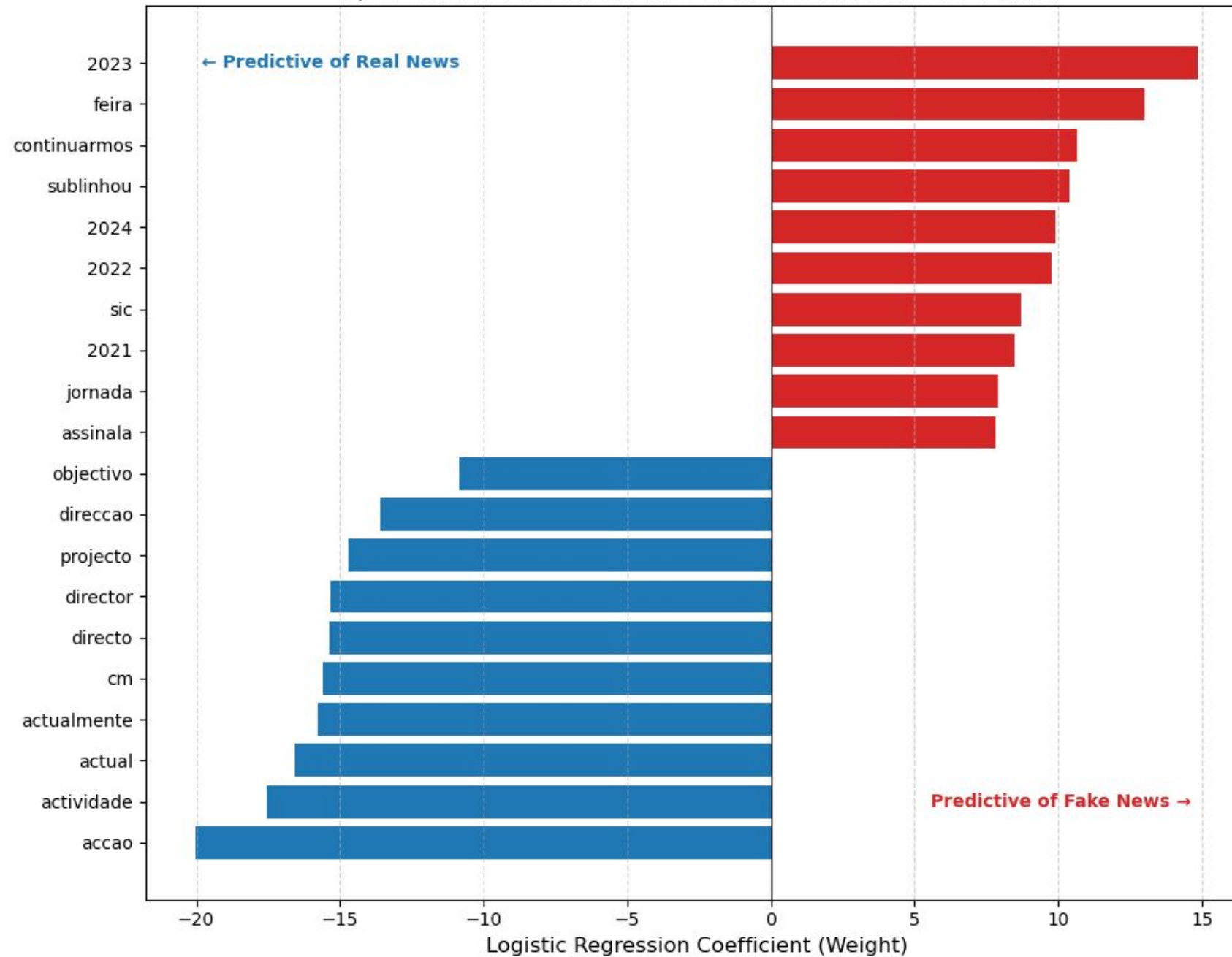


Prediction Probability Distribution - Best Model

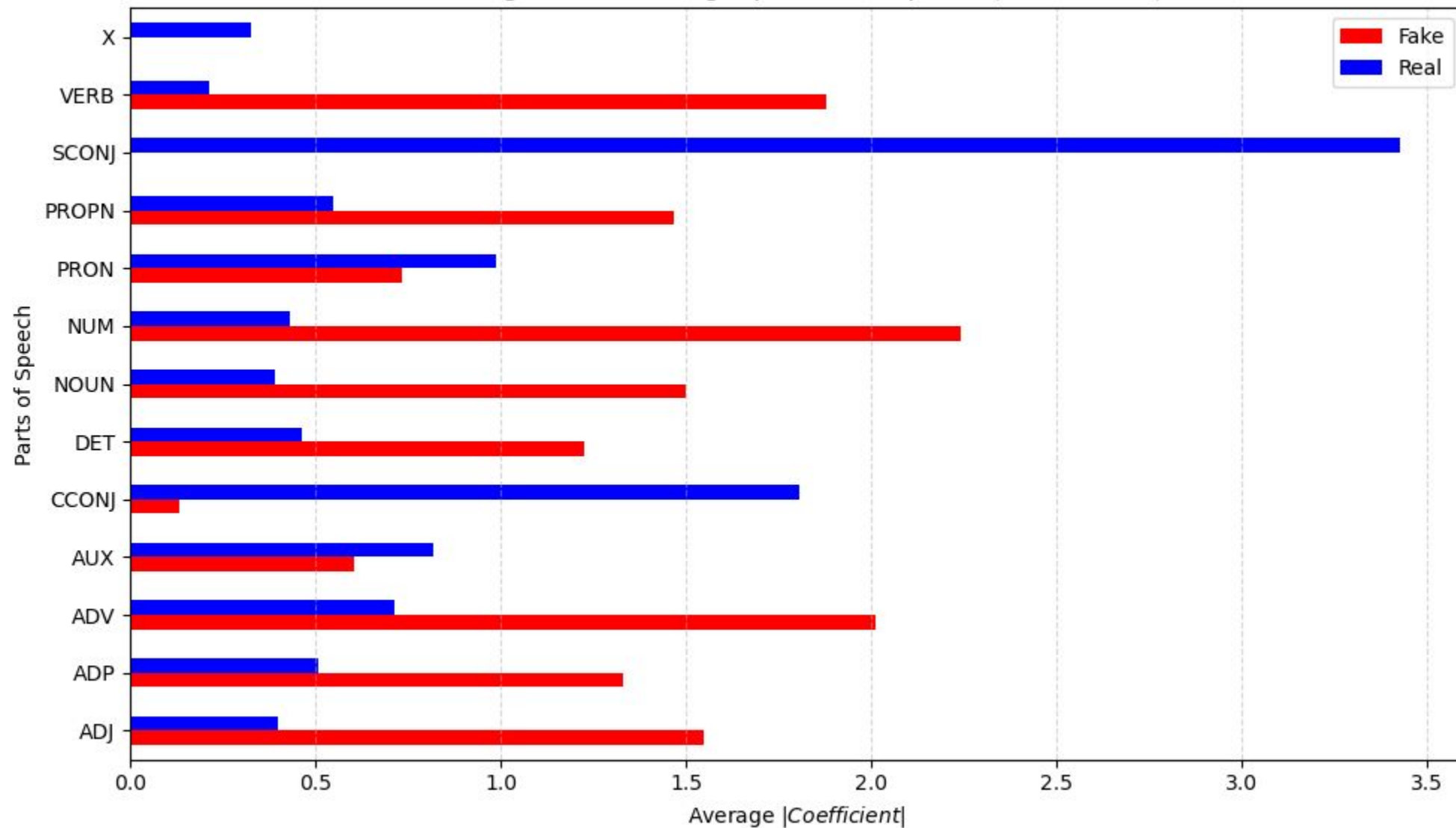


MODEL INTERPRETATION

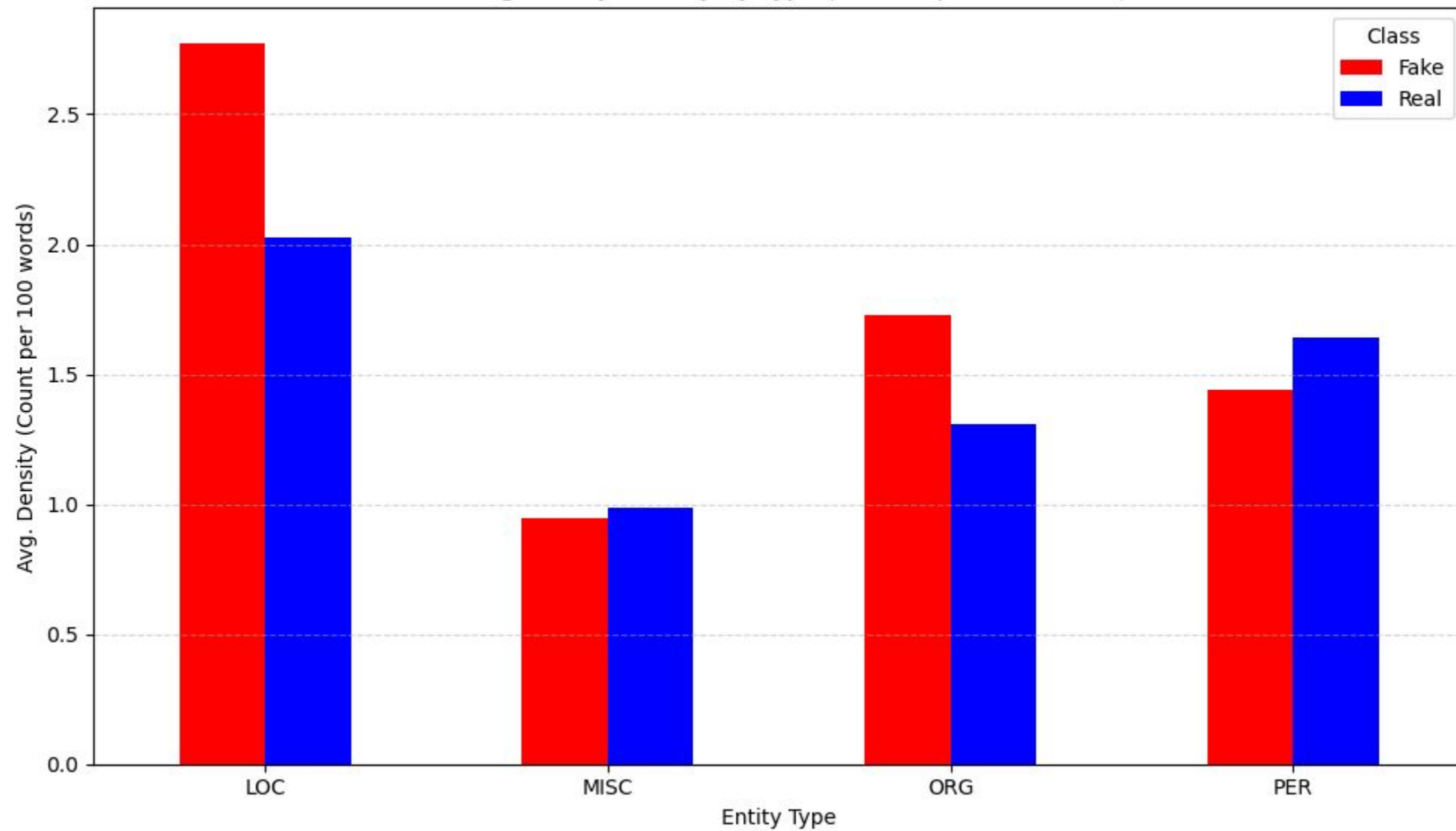
Top 10 Words Most Indicative of Real vs. Fake News (LR-L1)



Average Absolute Weight per Part of Speech (Fake vs Real)



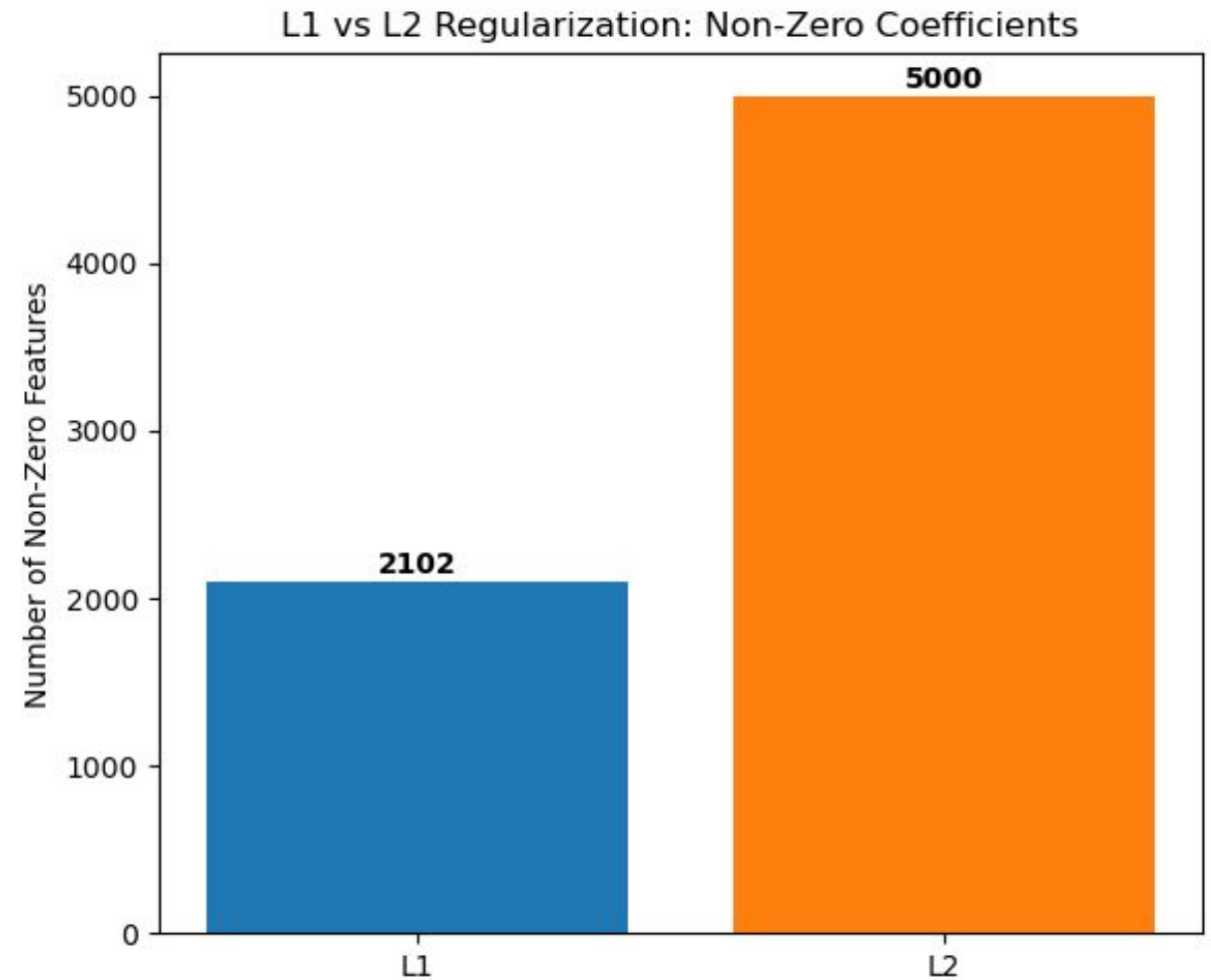
Average Entity Density by Type (Entities per 100 Words)

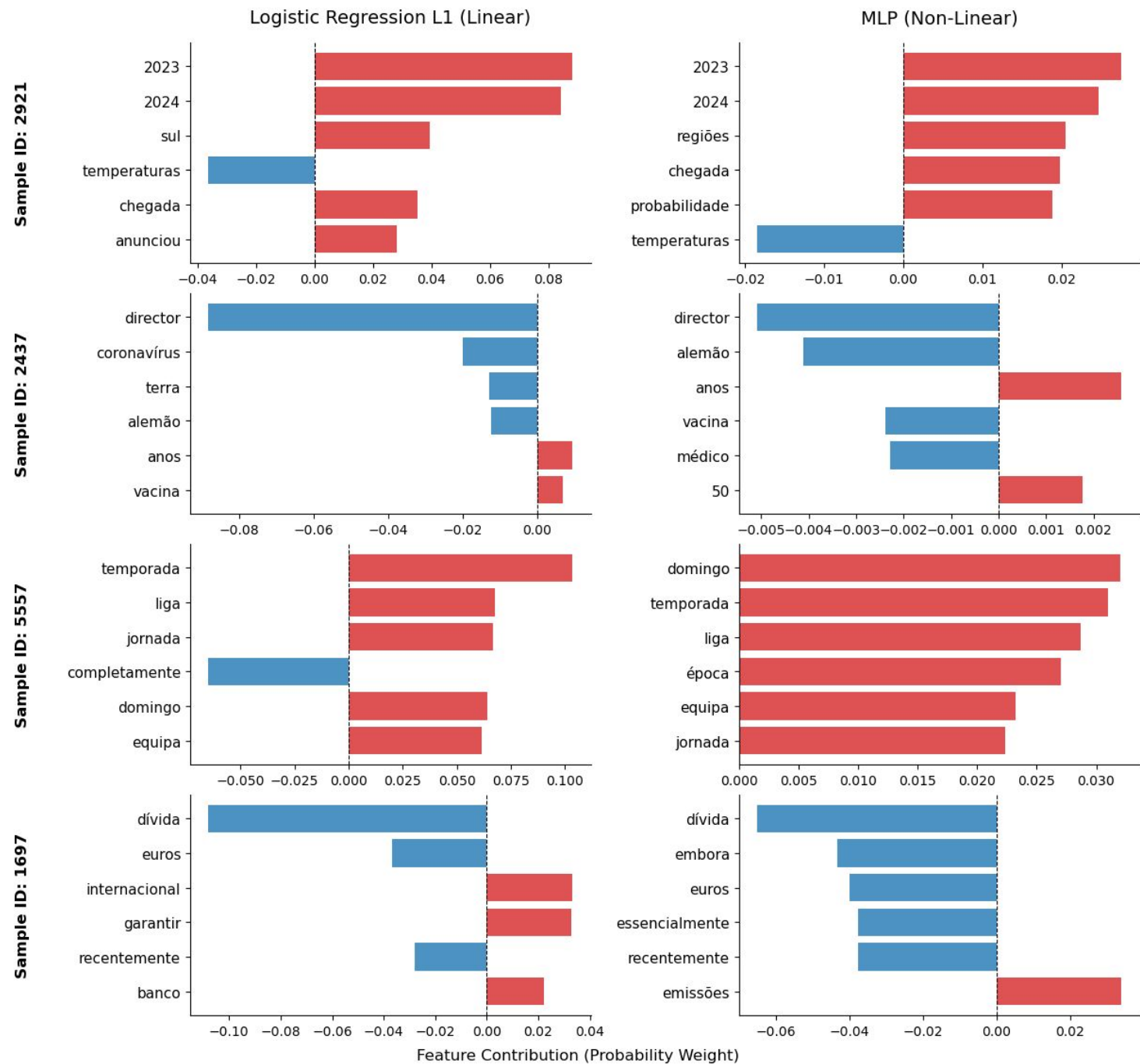


Regularization Comparison: L1 vs. L2

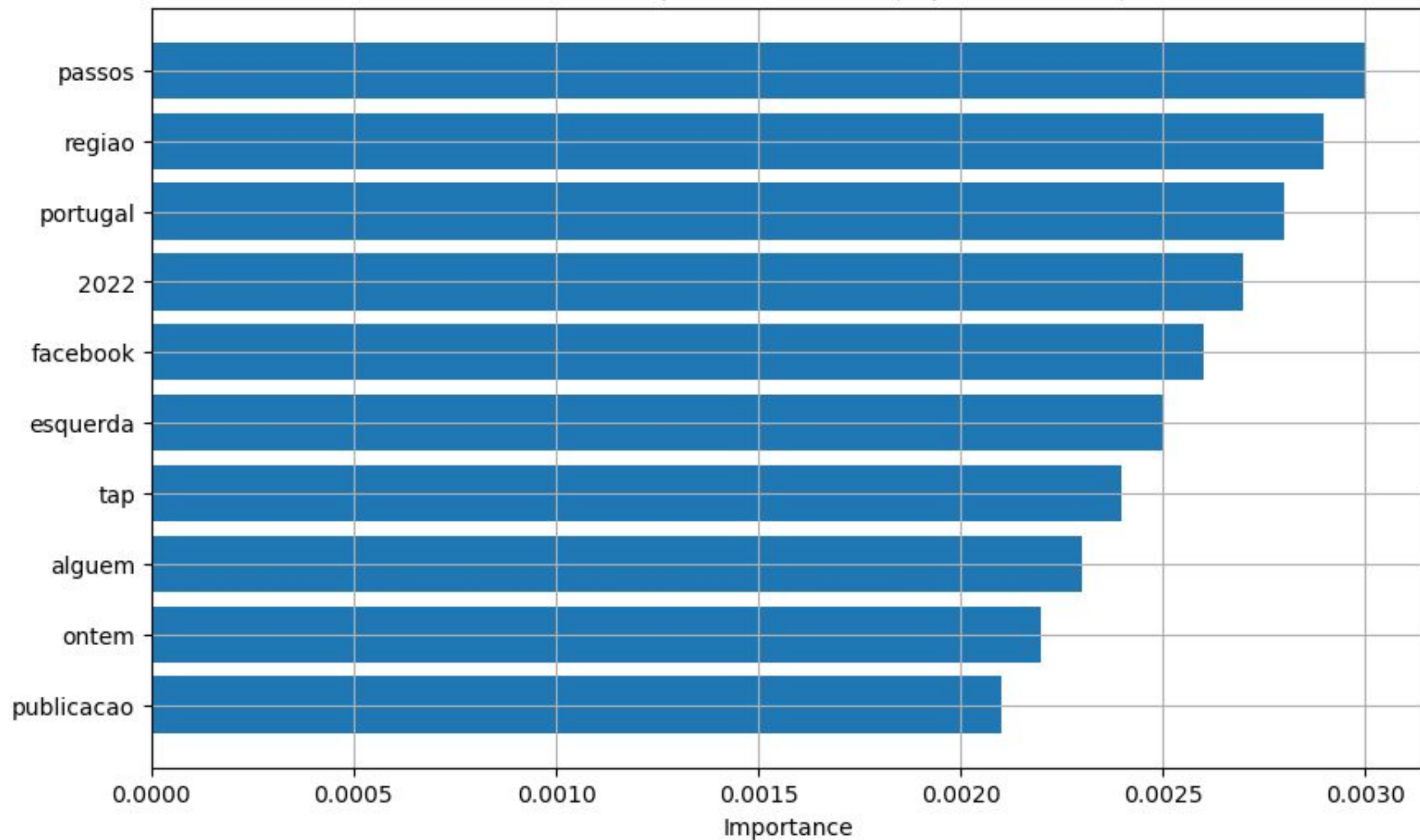
Which is more suitable?

- L1: Best for high-dimensional, noisy data where only a few features actually matter.
- L2: Best for highly correlated features where every variable contributes a little bit.



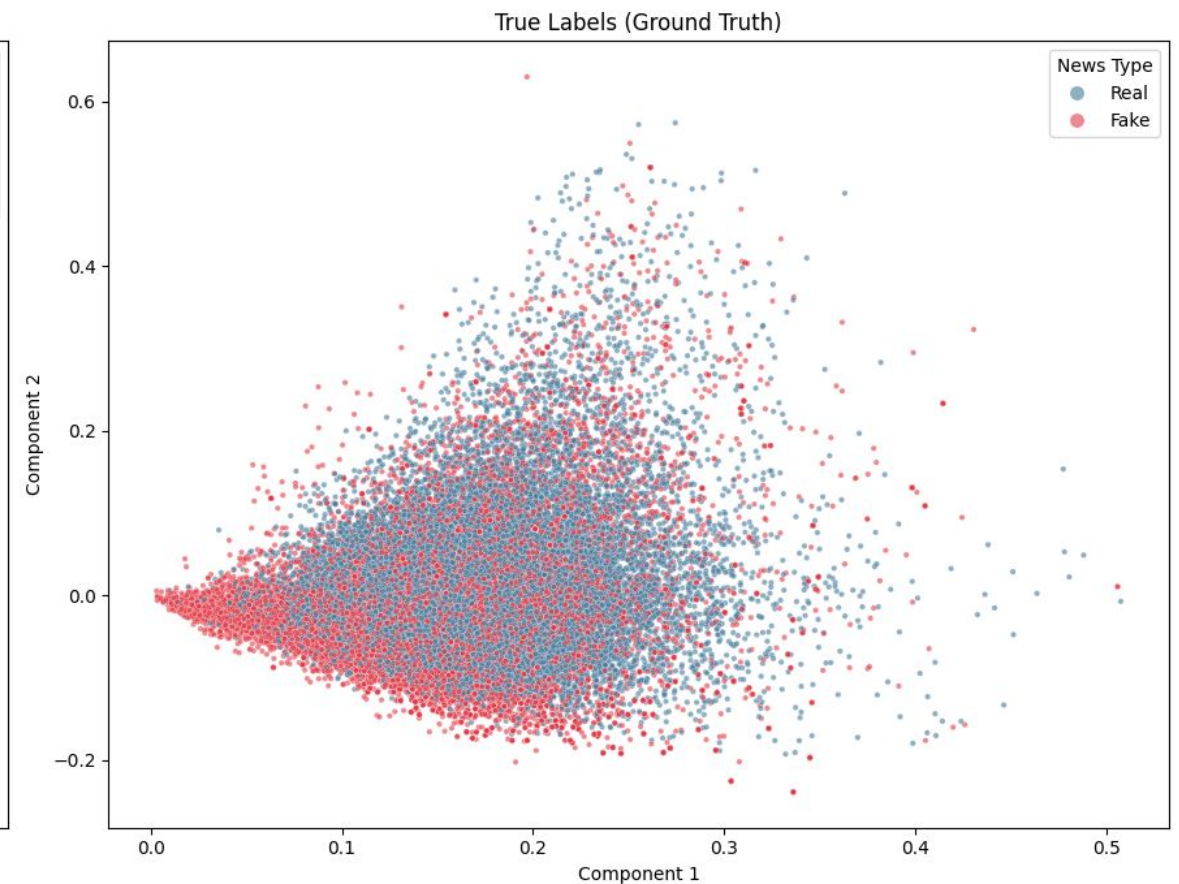
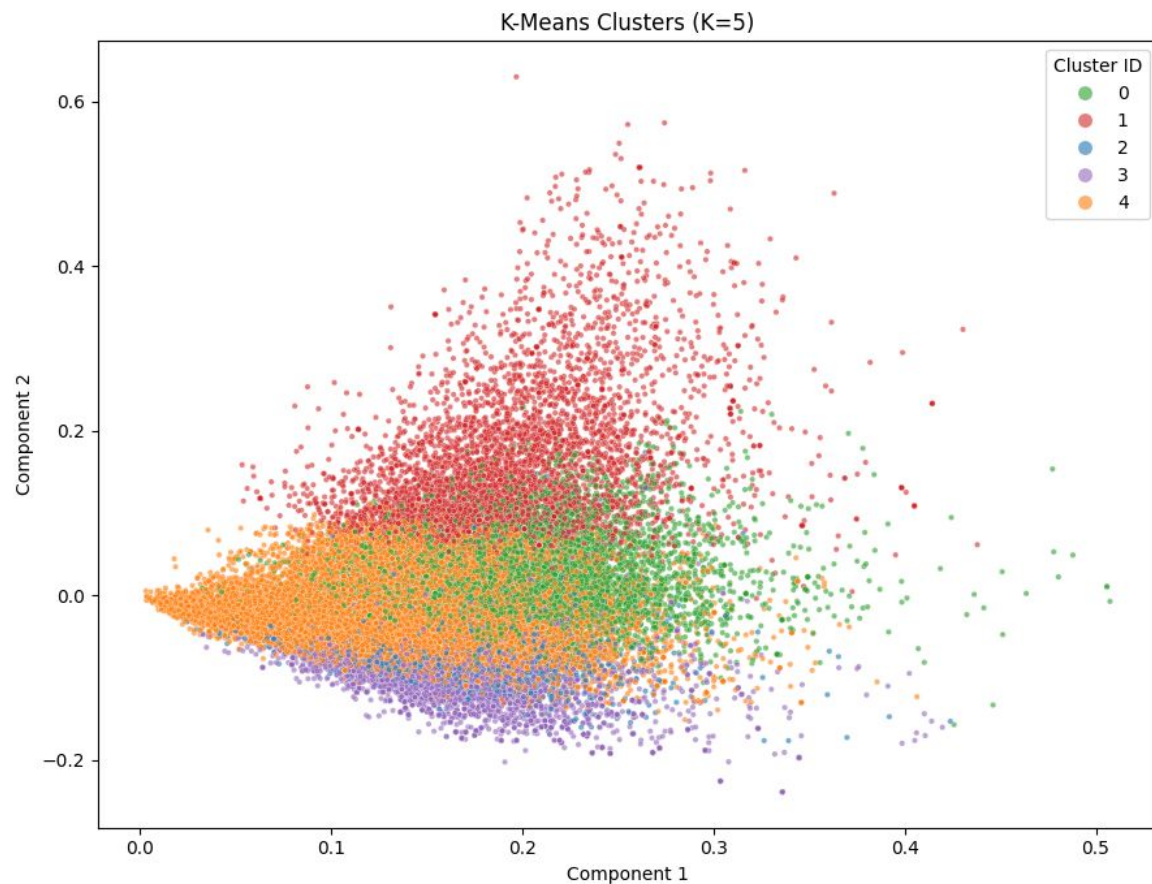


Permutation Importance — MLP (Top 10 Features)



Clustering

Label & Key Terms	Representative Snippets
Domestic Politics <i>governo, ministro, costa, ps, antonio, psd</i>	Ana Gomes: "Este governo também está a fabricar populismos". Militante de base, Ana Gomes é uma das vozes independentes...
Economy & Finance <i>euros, milhoes, mil, ano, portugal, valor</i>	Governo gasta dois mil milhões, mas tem margem para ir até 3,5 mil milhões. IVA a zero em alimentos essenciais...
Geopolitics (Ukraine War) <i>ucrania, russia, russo, guerra, putin, wagner</i>	Líder do Grupo Wagner quer batalha final para clarificar situação na Ucrânia: "Em teoria, a Rússia já pôs um fim a isto"...
Curiosities & Clickbait <i>anos, mae, jovem, homem, mulher, vida</i>	5 casos curiosos de crianças que afirmam se lembrar de suas vidas passadas. O que acontece quando morremos?...
Public Health & Society <i>pessoas, saude, portugal, anos, dia, ainda</i>	Manuel Carmo Gomes. "Não nos vamos ver livres da covid por várias razões". "O mais provável é que venhamos a recomendar a vacinação..."



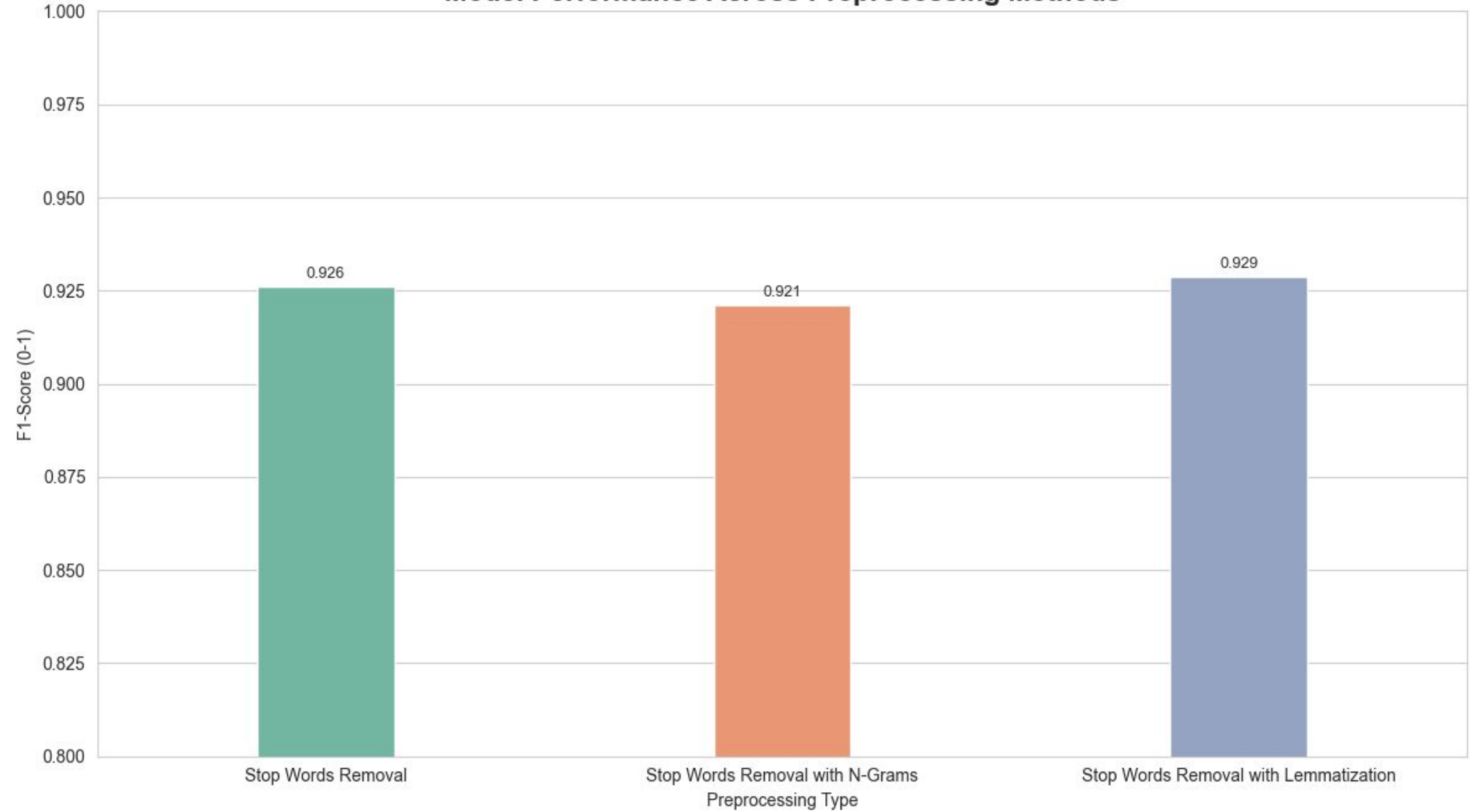
Label	 Domestic Politics	 Economy & Finance	 Geopolitics	 Curiosities & Clickbait	 Public Health & Society
Total Articles	8299	6096	2197	5791	28161
Fake News (%)	46.31%	49.72%	5.87%	76.53%	48.88%

Preprocessing Experiments

EXPERIMENT	HOW	REASON
N-Grams	Added N-Grams with range (1,2) to TF-IDF.	Standard unigrams (1,1) lose the relationship between adjacent words.
Lemmatization	Lemmatized the dataset with pt_core_news_lg[3] model.	Transforms inflected verbs and nouns into their dictionary base forms.

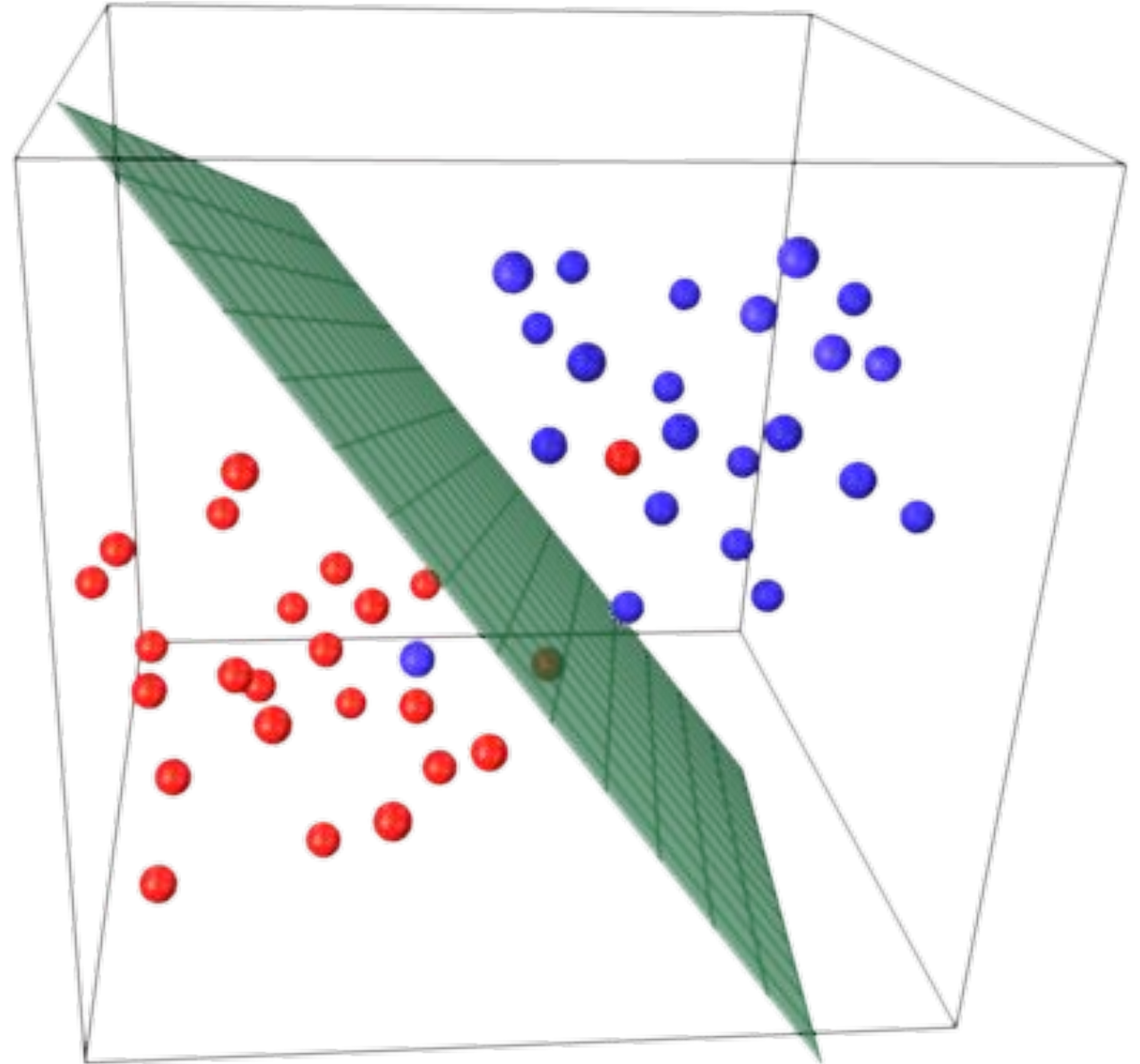
[3] : Portuguese Large Model trained on News dataset from spacy library

Model Performance Across Preprocessing Methods



SVM

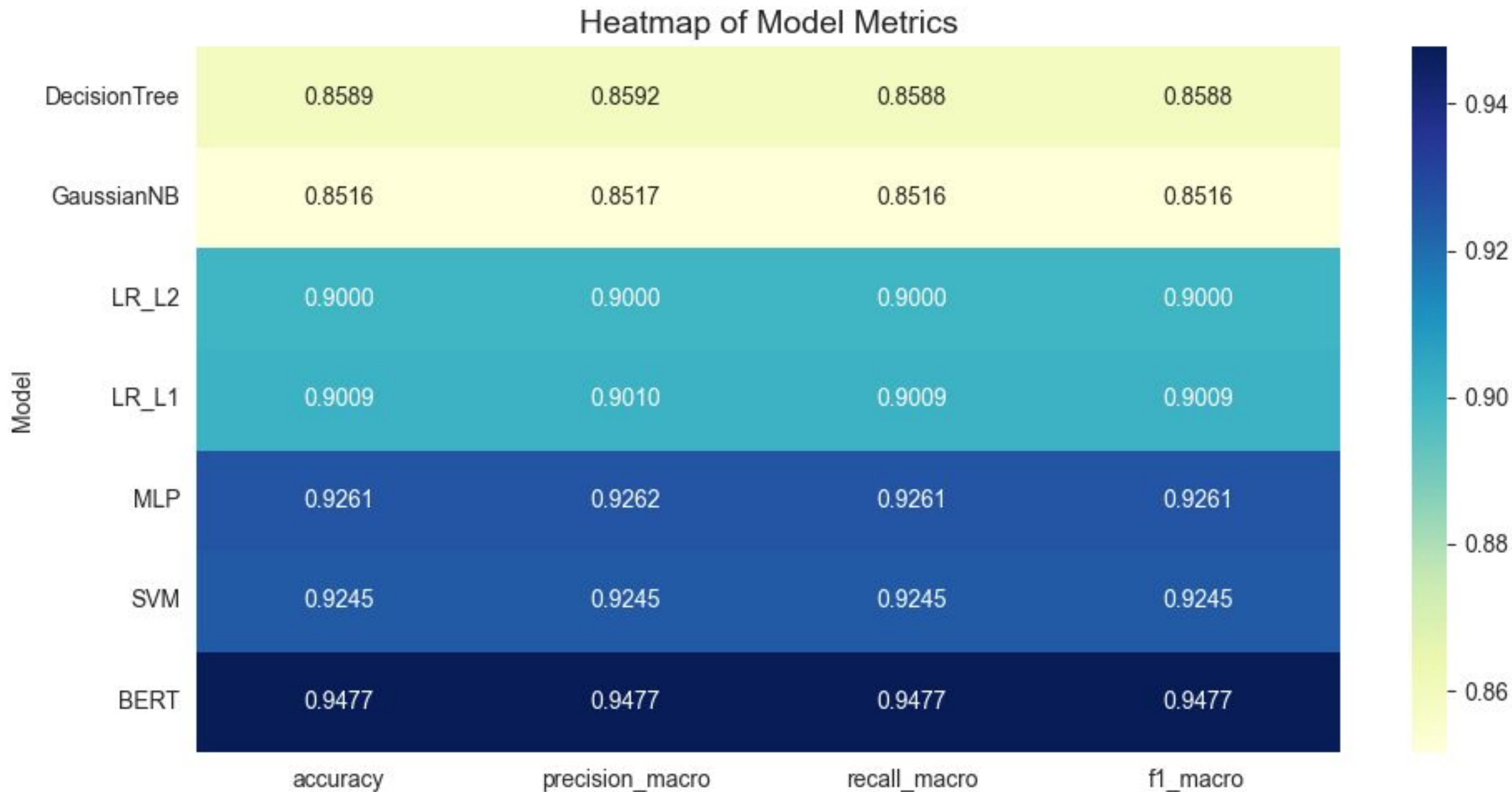
- Finds the optimal boundary between classes
- Allows some points to cross the line to handle messy data
- Focuses on maximizing the main space between categories



BERT

- Reads text in both directions to capture the full context
- Focuses attention on how specific words relate to each other
- Leverages deep language understanding to detect subtle patterns
- BERTimbau^[4]: A Pre-trained BERT model for Brazilian Portuguese by NeuralMind

[4]: <https://huggingface.co/neuralmind/bert-base-portuguese-cased>





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